

Lab 6

a. Using HTML and CSS, create a bouncing loader animation.

TAGS USED	ATTRIBUTES USED	DESCRIPTION
<!DOCTYPE html>	None	Declares that the document is an HTML5 document.
<html>	lang="en"	Root element of the HTML page and specifies English as the document language.
<head>	None	Contains metadata, the page title, and CSS styling.
<meta>	charset="UTF-8"	Defines the character encoding of the webpage.
<meta>	name="viewport"; content="width=device-width, initial-scale=1.0"	Makes the webpage adapt to different screen sizes.
<title>	None	Sets the title displayed on the browser tab.
<style>	None	Contains the CSS code used to style and animate the loader.
<body>	None	Contains the visible content of the webpage.
<div>	class="loader"	Creates the main container that holds the three animated shapes.
<div>	None	Each empty <div> represents one animated loader shape.

.loader	display: flex	Arranges the three shapes in a row using Flexbox.
.loader	justify-content: center	Centers the shapes horizontally inside the loader.
.loader	align-items: center	Centers the shapes vertically inside the loader.
.loader	gap: 20px	Creates equal spacing between the three shapes.
.loader	width: 180px	Sets the width of the loader container.
.loader	height: 120px	Sets the height of the loader container.
Body	display: flex	Uses Flexbox to position the loader on the page.
Body	justify-content: center	Centers the loader horizontally.
Body	align-items: center	Centers the loader vertically.
Body	min-height: 100vh	Makes the body at least as tall as the browser window.
background-color	#f5f7fa	Gives the webpage a light background color.
.loader div	width: 28px	Sets the width of each animated shape.
.loader div	height: 28px	Sets the height of each animated shape.
background	linear-gradient(...)	Gives the shapes a blue gradient appearance.

border-radius	50%	Initially makes the shapes circular.
box-shadow	0 6px 15px rgba(...)	Adds a soft shadow/glow beneath the shapes.
animation	bounceShape 2.4s ease-in-out infinite	Applies the shape-changing and bouncing animation continuously.
nth-child(2)	animation-delay: 0.2s	Delays the animation of the second shape by 0.2 seconds.
nth-child(3)	animation-delay: 0.4s	Delays the animation of the third shape by 0.4 seconds.
@keyframes	bounceShape	Defines the complete sequence of bouncing and shape-changing animation.
0%	transform, border-radius	Starting point of the animation where the shape begins as a circle.
12.5%	translateY(-45px), scale(1.05)	Moves the shape upward and slightly enlarges it.
25%	border-radius: 5%	Changes the circle into a square-like shape.
37.5%	rotate(45deg)	Rotates the square during the next bounce.
50%	rotate(45deg)	Produces the diamond-shaped appearance.
62.5%	rotate(90deg), border-radius: 35%	Rotates and rounds the shape during the next stage.
75%	border-radius: 35%	Maintains the rounded shape.

87.5%	rotate(180deg), border-radius: 50%	Rotates the shape and begins returning it to a circular shape.
100%	border-radius: 50%	Returns the shape to a circle and completes one animation cycle.
transform	translateY()	Moves the shape vertically to create the bouncing effect.
transform	rotate()	Rotates the shape to create different orientations such as the diamond effect.
transform	scale()	Slightly enlarges or reduces the shape during the animation.
border-radius	50%, 5%, 35%	Changes the appearance from circle to square-like and rounded shapes.
clip-path	polygon(...)	Defines a custom geometric shape by specifying its points.
Infinite	Animation value	Makes the animation repeat continuously.
ease-in-out	Animation timing function	Makes the animation start and finish smoothly.
linear-gradient	135deg, color values	Creates a gradual transition between the two blue colors.
rgba()	Color + transparency values	Creates semi-transparent shadow colors.
transform: translateY(-45px)	-45px	Moves a shape 45 pixels upward.

animation-delay	0.2s, 0.4s	Causes the three shapes to animate one after another instead of simultaneously.
alternate	Animation direction	In the earlier loader version, this made the animation reverse automatically; it is not used in the latest shape-changing code.

Final Outcome:



Code Snippet:

```
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport"
    content="width=device-width, initial-scale=1.0">

  <title>Bouncing Loader</title>

  <style>

    /* =====
    PAGE
    ===== */

    body {

      margin: 0;

      min-height: 100vh;

      display: flex;

      justify-content: center;

      align-items: center;

      background-color: #f5f7fa;

    }

    /* =====
    LOADER
    ===== */
```

```
.loader {  
  
    display: flex;  
  
    justify-content: center;  
  
    align-items: center;  
  
    gap: 20px;  
  
    width: 180px;  
  
    height: 120px;  
  
}  
  
/* =====  
  LOADER DOTS  
===== */  
  
.loader div {  
  
    width: 28px;  
  
    height: 28px;  
  
    background: linear-gradient(  
        135deg,  
        #4a90e2,  
        #6ec6ff  
    );  
  
    border-radius: 50%;  
  
    box-shadow:  
        0 6px 15px rgba(74, 144, 226, 0.30);  
  
    animation:  
        bounceShape 2.4s ease-in-out infinite;  
  
}
```

```
/* =====  
DIFFERENT START TIMES  
===== */
```

```
.loader div:nth-child(2) {  
  
    animation-delay: 0.2s;  
  
}
```

```
.loader div:nth-child(3) {  
  
    animation-delay: 0.4s;  
  
}
```

```
/* =====  
BOUNCE + SHAPE CHANGE  
  
0% - Circle  
25% - Square  
50% - Diamond  
75% - Rounded square  
100% - Circle  
===== */
```

```
@keyframes bounceShape {
```

```
    /* ROUND 1 - CIRCLE */
```

```
    0% {
```

```
        transform:  
            translateY(0)  
            rotate(0deg)  
            scale(1);
```

```
        border-radius: 50%;
```

```
        clip-path: polygon(
```

```
50% 0%,
100% 50%,
50% 100%,
0% 50%
);

}

/* Move upward + become square */
12.5% {

transform:
  translateY(-45px)
  rotate(0deg)
  scale(1.05);

border-radius: 20%;

clip-path: none;

}

/* ROUND 2 - SQUARE */
25% {

transform:
  translateY(0)
  rotate(0deg)
  scale(1);

border-radius: 5%;

clip-path: none;

}

/* Move upward + turn */
37.5% {
```

```
transform:
  translateY(-45px)
  rotate(45deg)
  scale(1.05);

border-radius: 10%;

}

/* ROUND 3 - DIAMOND */

50% {

  transform:
    translateY(0)
    rotate(45deg)
    scale(1);

  border-radius: 5%;

}

/* Move upward + become rounded */

62.5% {

  transform:
    translateY(-45px)
    rotate(90deg)
    scale(1.05);

  border-radius: 35%;

}

/* ROUND 4 - ROUNDED SHAPE */

75% {

  transform:
```

```
        translateY(0)
        rotate(90deg)
        scale(1);

border-radius: 35%;

}

/* Move upward + return to circle */

87.5% {

    transform:
        translateY(-45px)
        rotate(180deg)
        scale(1.05);

border-radius: 50%;

}

/* BACK TO CIRCLE */

100% {

    transform:
        translateY(0)
        rotate(180deg)
        scale(1);

border-radius: 50%;

}

}

</style>

</head>

<body>
```

```
<!-- =====
  SHAPE CHANGING LOADER
===== -->

<div class="loader">

  <div></div>

  <div></div>

  <div></div>

</div>

</body>

</html>
```

b. Create a photo gallery with CSS Flex.

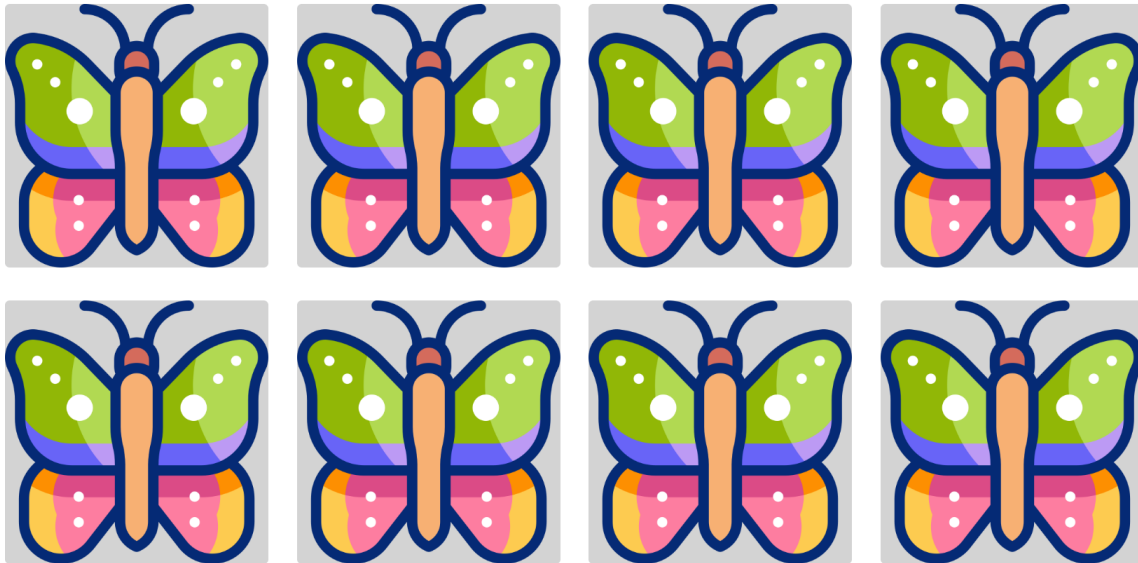
TAGS USED	ATTRIBUTES USED	DESCRIPTION
<!DOCTYPE html>	None	Declares that the document uses HTML5.
<html>	lang="en"	Root element of the HTML document and specifies English as the language.
<head>	None	Contains metadata, viewport settings, title, and CSS styling.
<meta>	charset="UTF-8"	Defines the character encoding of the webpage.
<meta>	name="viewport"; content="width=device-width, initial-scale=1.0"	Makes the webpage responsive by adapting its width to the device screen.
<title>	None	Sets the title of the webpage shown in the browser tab.
<style>	None	Contains the internal CSS used to

		design and make the gallery responsive.
<body>	None	Contains the visible content of the webpage.
<h2>	None	Displays the heading "Photo Gallery".
<div>	class="gallery"	Creates the main gallery container that holds all the photos.
<div>	class="photo"	Creates an individual container for each photo.
	src="butterfly_3.png"	Specifies the image file to be displayed.
	alt="Photo 1" etc.	Provides alternative text for the image if it cannot be displayed and improves accessibility.
.gallery	display: flex	Makes the gallery a Flexbox container.
.gallery	gap: 30px	Creates 30px of space between the photos.
.gallery	flex-wrap: wrap	Allows photos to move to the next line when there is not enough horizontal space.
.gallery	justify-content: center	Centers the photos horizontally inside the gallery.
.photo	width: 20%	Sets each photo container to 20% of the available width on larger screens.
.photo img	width: 100%	Makes each image fill the complete width of its photo container.
.photo img	background-color: lightgray	Gives a light-gray background to the image element.
.photo img	border-radius: 5px	Rounds the corners of the images.
.photo img:hover	transform: scale(1.05)	Enlarges the image slightly when the mouse pointer is moved over it.
:hover	scale(1.05)	Creates a zoom-in hover effect on the image.
@media	max-width: 700px	Creates a responsive layout for screens 700px wide or smaller.
.photo inside	width: 40%	Increases each photo width to

@media		40% on smaller screens, allowing fewer photos per row.
@media	max-width: 400px	Creates another responsive layout for very small screens.
.photo inside @media	width: 80%	Makes each photo occupy 80% of the available width on very small screens.
display: flex	Flexbox	Arranges the images in a flexible row-based layout.
flex-wrap: wrap	wrap	Allows the images to automatically move to new rows.
justify-content	center	Centers the items horizontally.
gap	30px	Controls the spacing between gallery items.
transform	scale(1.05)	Creates the zoom effect when hovering over an image.
transition	Not used in your original code	The original code does not define a transition, so the zoom happens immediately.
@media	max-width: 700px, max-width: 400px	Makes the photo gallery responsive for tablets and mobile devices.

Final Outcome:

Photo Gallery



Code Snippet:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <style>

  h2{
    text-align: center;
  }
  .photo
  {
    width:20%;
  }
  .photo img{
    width:100%;
    background-color:lightgray;
    border-radius: 5px;
```

```
}
.photo img:hover
{
  transform: scale(1.05);
}
.gallery
{
  display: flex;
  gap: 30px;
  flex-wrap: wrap;
  justify-content: center;
}
@media (max-width: 700px) {
  .photo
  {
    width: 40%;
  }
}
@media (max-width: 400px) {
  .photo
  {
    width: 80%;
  }
}
</style>
</head>
<body>
<h2>Photo Gallery</h2>
<div class="gallery">

  <div class="photo"></div>
  <div class="photo"></div>
  <div class="photo"></div>
  <div class="photo"></div>
  <div class="photo"></div>
  <div class="photo"></div>
  <div class="photo"></div>
  <div class="photo"></div>

</div>
```

```
</body>
</html>
```

c. Apply the Flex property on flex container and flex child.

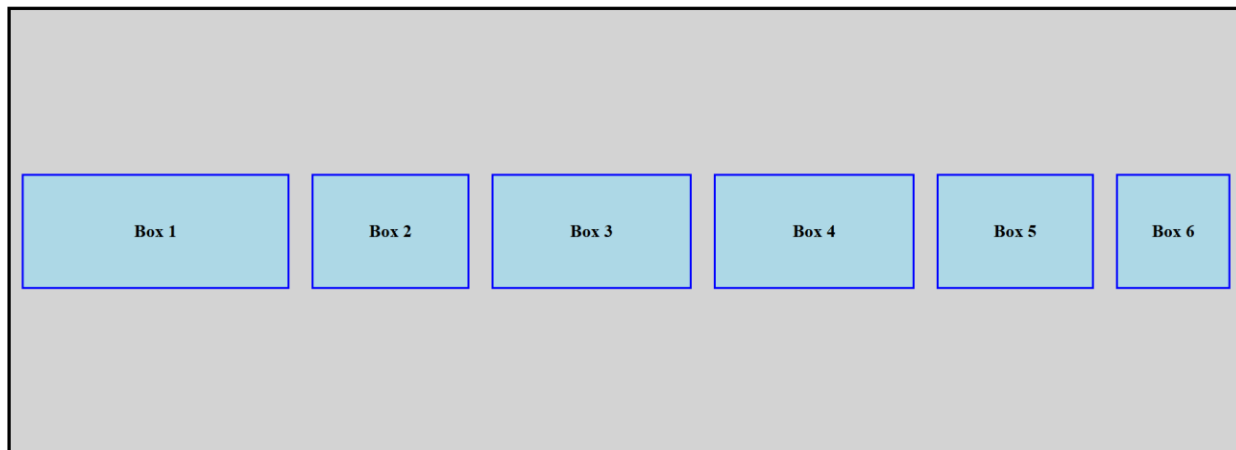
TAGS USED	ATTRIBUTES USED	DESCRIPTION
<!DOCTYPE html>	None	Declares the document as an HTML5 document.
<html>	lang="en"	Root element of the HTML document and specifies English as the language.
<head>	None	Contains metadata, title, and CSS styling.
<meta>	charset="UTF-8"	Defines the character encoding used by the webpage.
<meta>	name="viewport"; content="width=device-width, initial-scale=1.0"	Makes the webpage adapt to different screen sizes.
<title>	None	Sets the title shown in the browser tab.
<style>	None	Contains the internal CSS used to style the Flexbox examples.
<body>	None	Contains the visible webpage content.
<h2>	None	Displays the heading of the webpage.
<div>	class="container"	Creates the Flex container that holds the child elements.
<div>	class="box"	Defines a common style for the Flex child elements.
<div>	class="box box1" etc.	Applies the common .box styling and individual child properties such as .box1, .box2, etc.
border	2px solid blue / 3px solid black	Adds a visible border around boxes or the Flex container.
margin	10px	Creates space around the boxes.
background-color	lightblue, lightgrey	Sets the background color of boxes and the container.
text-align	center	Aligns the text horizontally in the center.
font-weight	bold	Makes the text bold.
display	flex	Makes the .container a Flex container.

flex-direction	row	Places Flex children horizontally from left to right.
justify-content	flex-start	Places Flex items at the beginning of the main axis.
align-items	flex-start	Aligns Flex items at the beginning of the cross axis.
flex-wrap	wrap	Allows Flex items to move onto multiple lines when there is not enough space.
align-content	space-around	Distributes multiple rows/lines of Flex items with equal space around them.
height	300px, 400px	Sets the height of the Flex container.
width	100px, 100%-related available space	Sets the width of the first container or the width of individual boxes.
line-height	100px	Vertically centers the text inside a 100px-high box.
flex-grow	1, 2	Controls how much a Flex child can grow to use available extra space.
flex-basis	200px	Sets the initial main-axis size of a Flex child before extra space is distributed.
flex-shrink	10, 9	Controls how much a Flex child shrinks when there is not enough space. A larger value means greater shrinking relative to items with smaller shrink factors.
align-self	flex-start	Overrides the container's align-items setting for an individual Flex child.
.container	display: flex	Defines the parent as the Flex container.
.box	Common child styling	Defines the basic appearance shared by all Flex child elements.
.box1	flex-grow: 1	Allows Box 1 to grow and also gives it an initial size of 200px through flex-basis; its flex-shrink: 10 gives it a high shrink factor.
.box2	flex-grow: 1, align-self: flex-start	Allows Box 2 to grow and individually aligns it to the start of the cross axis.
.box3	flex-grow: 2	Allows Box 3 to take twice the growth factor of a child with flex-grow: 1 when extra space is distributed.

.box4	flex-grow: 2	Gives Box 4 a growth factor of 2.
.box5	flex-grow: 1	Gives Box 5 a growth factor of 1.
.box6	flex-shrink: 9	Gives Box 6 a high shrink factor when the container does not have enough space.
flex-grow	Child property	Determines how a flex item grows when extra space is available.
flex-shrink	Child property	Determines how a flex item shrinks when available space is insufficient.
flex-basis	Child property	Defines the initial size of a flex item before growing or shrinking.
align-self	Child property	Allows one particular flex child to have a different cross-axis alignment from the other children.

Final Outcome:

Flex Container and Flex Child Properties



Code Snippet:

```
<!DOCTYPE html>
<html lang="en">

<head>
```

```
<meta charset="UTF-8">

<meta name="viewport"
  content="width=device-width, initial-scale=1.0">

<title>Flex Container and Flex Child Properties</title>

<style>

  /* =====
  FLEX PARENT / FLEX CONTAINER
  ===== */

  .container {

    display: flex;

    flex-direction: row;

    justify-content: flex-start;

    align-items: flex-start;

    flex-wrap: wrap;

    align-content: space-around;

    background-color: lightgrey;

    height: 400px;

    width: 100%;

    border: 3px solid black;

  }

  /* =====
  COMMON FLEX CHILD
  ===== */
```

```
.box {  
  
    width: 100px;  
  
    height: 100px;  
  
    margin: 10px;  
  
    background-color: lightblue;  
  
    border: 2px solid blue;  
  
    text-align: center;  
  
    line-height: 100px;  
  
    font-weight: bold;  
  
}  
  
/* =====  
   FLEX CHILD 1  
   ===== */  
  
.box1 {  
  
    flex-grow: 1;  
  
    flex-basis: 200px;  
  
    flex-shrink: 10;  
  
}  
  
/* =====  
   FLEX CHILD 2  
   ===== */  
  
.box2 {
```

```
flex-grow: 1;

align-self: flex-start;
}

/* =====
FLEX CHILD 3
===== */

.box3 {

flex-grow: 2;
}

/* =====
FLEX CHILD 4
===== */

.box4 {

flex-grow: 2;
}

/* =====
FLEX CHILD 5
===== */

.box5 {

flex-grow: 1;
}

/* =====
FLEX CHILD 6
===== */

.box6 {
```

```
        flex-shrink: 9;

    }

</style>

</head>

<body>

    <h2>Flex Container and Flex Child Properties</h2>

    <!-- =====
        FLEX PARENT
    ===== -->

    <div class="container">

        <!-- FLEX CHILDREN -->

        <div class="box box1">
            Box 1
        </div>

        <div class="box box2">
            Box 2
        </div>

        <div class="box box3">
            Box 3
        </div>

        <div class="box box4">
            Box 4
        </div>

        <div class="box box5">
            Box 5
        </div>

    </div>

</body>

</html>
```

```
<div class="box box6">  
  Box 6  
</div>  
  
</div>  
  
</body>  
  
</html>
```